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Effect of number of blades on aerodynamic forces on a straight-bladed Vertical Axis Wind Turbine

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ABSTRACT

Small wind turbine performance and safety standard for straight-bladed Vertical Axis Wind Turbine (VAWT) have not been developed in the world because of the lack of fundament experimental data. This paper focuses on the evaluation of aerodynamic forces depending on several numbers of blades in wind tunnel experiment. In the present study, the test airfoil of blade is symmetry airfoil of NACA 0021 and the number of blades is from two to five. Pressure acting on the surface of rotor blade is measured during rotation by multiport pressure devices and transmitted to a stationary system through wireless LAN. And then, the aerodynamic forces (tangential force, normal force et al.) are discussed as a function of azimuth angle, achieving a quantitative analysis of the effect of numbers of blades. Finally, the loads are compared with the experimental data of six-component balance. As a result, it is clarified that the power coefficient decreases with the increase of numbers of blades. Furthermore, the power which is absorbed from wind by wind turbine mainly depends on upstream region of azimuth angle of $\theta = 0°$~$180°$. In this way, these results are very important for developing the simple design equations and applications for straight-bladed VAWT.

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1. Introduction

The wind energy has been one of the veritable alternative resources for power production in recent years because of the potential that they offer for carbon free power generation [1,2,33]. Wind energy technology development is mainly attributing of HAWTs (Horizontal Axis Wind Turbines) which have major recognized implementation, such as large scale windfarms on flat land, offshore and mountainous terrain [4]. For VAWTs (Vertical Axis Wind Turbines), the generator can be placed near the ground, so the tower does not need to support it, also makes maintenance easier wind and always be used in urban areas, as shown in Fig. 1. This type VAWT can reduce the transmission losses due to proximity to the demand center. Moreover, VAWT can be applied in the remote areas, street lights and general families, etc, because of its independent power generation system. It also can provide power for portable device, crisis evacuation indicator in disaster events, etc. Therefore, there has been an increasing interest in the deploying VAWTs in urban areas [5]. However, compared with the HAWT, very few VAWTs are available commercially [3].

According to IEC (International Electrotechnical Commission), AWEA (American Wind Energy Association) and BWEA (British Wind Energy Association), small wind turbine performance and safety standard have been developed for HAWTs [6]. For example, IEC 61400-2 and JISCI1400-2 have a very good expression of simple design equations and design standards. However, R & D of the performance and safety standard for small VAWT have not been developed in the world because of the lack of fundament experimental data [7].

VAWTs are classified into Darrieus type wind turbine (Egg-beater type wind turbine, Straight-bladed type wind turbine, Crossflex wind turbine and so on), Savonius rotor, Combined Savonius and Darrieus rotor, Sistan type wind turbine and so on [8]. In this study, small straight-bladed VAWT which is the most popular in domestic and foreign is studied as the research object. Blades of VAWT are made of uniform section and non-twisted, making them relatively easy to fabricate and extrude.

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Nomenclature

A	swept area of wind turbine [m^2]
c	airfoil chord length (=0.265) [m]
C_D	drag coefficient ($=F_D/(0.5\rho cU_0^2)$)
C_L	lift coefficient ($=F_L/(0.5\rho\ cU_0^2)$)
C_p	pressure coefficient ($=P/(0.5\rho U_0^2)$)
C_{power}	power coefficient ($=Q\omega/(0.5\rho DHU_0^3)$)
C_Q	torque coefficient ($=Q/(0.5\rho DHRU_0^2)$)
C_N	normal force coefficient ($=F_N/(0.5\rho cU_0^2)$)
C_{thrust}	thrust coefficient ($=F_x/(0.5\rho DHU_0^2)$)
C_T	tangential force coefficient ($=F_T/(0.5\rho cU_0^2)$)
D	rotor diameter (=2.0) [m]
F_D	drag force per unit length [N/m]
F_L	lift force per unit length [N/m]
F_N	normal force per unit length [N/m]
F_T	tangential force per unit length [N/m]
F_x	thrust force per unit length [N/m]
H	height of rotor blade (=1.2) [m]
N	number of blades (2~5)
P	pressure on the surface of blade [Pa]
P_{ower}	power putout [W]
Q	rotor torque [N m]
R	rotor radius (=1.0) [m]
Re	local Reynolds number ($=Wc/\nu$)
U_0	mainstream wind velocity [m/s]
V	tip speed of blade ($=R\omega$) [m/s]
W	resultant flow velocity [m/s]
α	angle of attack [deg]
β	blade pitch angle [deg]
θ	azimuth angle [deg]
λ	tip speed ratio ($=R\omega/U_0$)
ν	kinematic viscosity [m^2/s]
ρ	air density [kg/m^3]
ω	angular velocity of rotor [rad/s]

In the development of the simple design equations for a HAWT, loads which are applied to blade and rotor shaft are mainly considered [9]. However, VAWT load is also applied to the support structure. To define the directions of the loads of straight-bladed VAWT, the systems of coordinate are shown in Fig. 2.

As mentioned above, the support structure coordinate system shown the different coordinate systems of blade and rotor shaft. Compared with the HAWT, the loads applied to VAWT become more and more complicated. For HAWT, centrifugal force is caused by the rotation. From this figure, it is seen clearly that blade is tensile in the radial direction [10]. Therefore, in the designing of VAWT, the tensile strength is mainly considered.

Fig. 3 illustrates blade load in the flap direction for straight-bladed VAWT. As shown in this figure, it is noted that not only the centrifugal loads are caused by the rotation, but also the bending loads are generated from fulcrum. Thence, in the designing of VAWT, the bending strength is mainly considered [11]. Furthermore, the flow field characteristics are also different between HAWT and VAWT [12]. Wind uniformly flows into the rotor surface and the torque constant for HAWT. But, for VAWT, wind velocity which flows into the rotor surface becomes disturbed flow in the downstream region [13]. Moreover, large fluctuation torque will be generated and fatigue loads of support structure become stringent.

Therefore, as described above, the simple design methods and standards of HAWT is not suitable for the load cases of VAWT. In the design process of VAWT it is crucial to maximize the aerodynamic performance ratio [14]. Therefore, industries and researchers are trying to focus aerodynamic performance for VAWTs with wind tunnel experimentally and numerically. Based on the development of the simple design equations for HAWT, it is very important to determine the parameters by experiences and theoretical [15]. The maximum power coefficient of VAWT mainly depends on rotor



Fig. 1. Small Vertical Axis Wind Turbine.

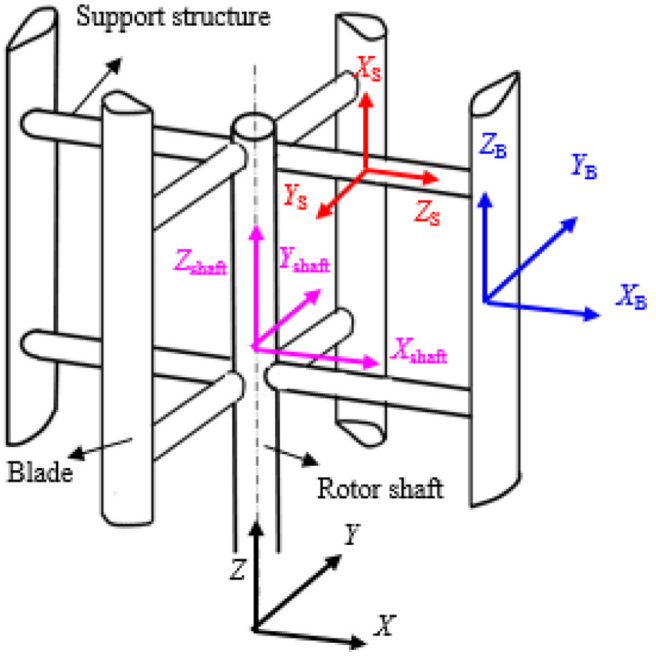


Fig. 2. Coordinate systems of Vertical Axis Wind Turbine.

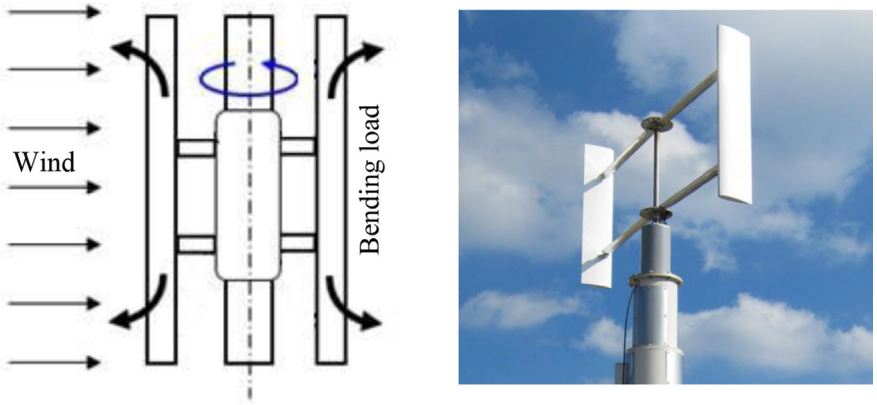


Fig. 3. Blade load in the flap direction of Vertical Axis Wind Turbines.

solidity, wind speed, tip speed ratio and so on [16]. For a VAWT rotor solidity depends on the number of blades, rotor radius and airfoil chord. Hence this paper focuses on the numbers of blade for straight-bladed VAWT.

In SH. et al. [17] presented a commercial CFD program analysis of energy performance and aerodynamic force on a four-bladed VAWT at different pitch angles. From this work, it can be seen that changing the pitch angle has a pronounced effect on power coefficient. However, the tip speed ratio at which peak power occurred changed slightly depending on the pitch angle. Their results were similar to investigations in Armstrong S. et al. [18], Staelens et al. [19] and Paraschivoiu I. et al. [20]. Moreover, Elkhoury M. et al. [4] examined the effects of wind speed, turbulence intensity and airfoil shape on the performance of wind turbine with experimentally and numerically. Wind velocity of 8 m/s or above would be rarely encountered, the effects of wind velocity was also studied by Tullis S. et al. [21] and Li QA. et al. [22]. Furthermore, it was found that the symmetric airfoil had a slightly better performance corresponding to maximum power coefficient. This work was also performed in Sun H. et al. [23], Li J. et al. [24], as well as Ismail MF. et al. [5].

In order to obtain a very detailed understanding of the main aspects of the dynamic stall phenomenon, Castelli MR. et al. [25] and Nobile R. et al. [26] focused on the phenomena of VAWT performance and aerodynamic forces at different tip speed ratios based on CFD. Also, the effect of dynamic stall was investigated by Simão Ferreira C. et al. [27], reporting the influence of the turbulence model in the simulation of the vortical structures spread from the blade itself. The obtained results shown the reduction of blade relative angles of attack passing from lower to higher tip speed ratio values, due to the increasing influence of blade translational speed in the near-blade flow field. Maeda T. et al. [28] clarified the velocity field and the pressure distribution around VAWT with two-blade VAWT by wind tunnel measurement. As this results, the geometrical angle of attack changes periodically due to local wind velocity and direction depending on azimuth angle. The tangential force and the drag acting on the blade show the maximum values when the blade passes upstream region. In this research, the aerodynamic performance and the flow physics surrounding VAWT blades in unsteady winds were investigated.

Between the many factors that influence the aerodynamic behavior of the rotor, an important role is played by its number of blades. Marco RC. et al. [29] discussed a CFD model for the evaluation of the torque variation during the revolution of VAWT by increasing the number of blades. As shown in this research, the peak of power coefficient decreased with the increase of number of blades, while it moved to lower tip speed ratio. The maximum torque values are generated during the upwind revolution of the turbine and for azimuthal positions where rotor blades are experiencing very high relative angles of attack. With regard to the radial component of the aerodynamic forces, an increase in number of blades brings to a decrease of this force, which is desirable from a structural perspective. Similar result has been investigated by Tangler JL. et al. [30], as well as Li S. et al. [31]. However, these recently accomplished studies do not account for support structure that tend to have considerable influence on the performance of VAWT. Moreover, for all those works were mainly carried out with CFD, rather than from the wind tunnel experiment. It is still very difficult to fully solve aerodynamic problem by CFD model.

So far, very little work has been carried out into the effects of the number of blades on the VAWT aerodynamic forces in wind tunnel experiment. Specifically, there have been few reports relevant to low tip speed ratio during rotation, because the flow field around VAWT rotor blade is so much complicated that the aerodynamic characteristics of the blades of a small straight-bladed VAWT become more and more complicated [32].

For detailed review of various configurations and design techniques of the VAWT, in this research, wind tunnel experiment is carried out. In order to eliminate the influence of the support structure and shaft, the aerodynamic characteristics of single blade at different numbers of blades are experimentally investigated through directly measurement of the load acting on the blade in wind tunnel. And then, the six-component balance which is installed on the basement of wind turbine can measure the force and the moment applied to the entire wind turbine, in the *x*, *y* and *z*-axis directions. Finally, the experimental results are compared with different numbers of blades. The effects of the azimuth angle, the tip speed ratio, and the number of blades on the performance of the VAWT are investigated. In this way, an accurate and complete measurement of the aerodynamic loading on the turbine blades could be obtained to better guide to the development of the simple design for straight-bladed VAWT.

2. Experimental procedure

2.1. Experimental apparatus

In wind tunnel, a test wind turbine is presented in this experiment to clear the effects of number of blades on the VAWT aerodynamic performance in two-dimensional. The contents of measurement are shown as following.

Fig. 4 indicates the schematic diagram of the whole experimental apparatus. As shown in this figure, pitot tube which is installed in 2.07 m of upstream from rotor shaft can determine

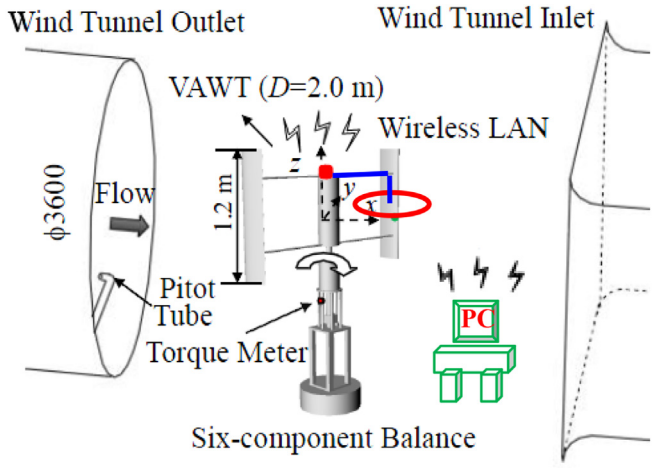


Fig. 4. Schematic diagram of experimental apparatus. Torque meter which is installed in wind turbine axis of rotation can determine rotational speed and rotor torque. Pitot tube can determine mainstream velocity.

mainstream wind velocity. And then, torque meter which is installed in rotor shaft of wind turbine can determine power and torque coefficients absorbed from wind turbine. In order to clear aerodynamic performance, multiport pressure devices which are installed on the rotor hub of wind turbine can measure the pressures acting on the rotor blade surface. Finally, the six-component balance which is installed on the basement of wind turbine can measure the force and the moment applied to the entire wind turbine, in the *x*, *y* and *z*-axis directions.

2.2. Wind tunnel and wind turbine

An open test section of circular type wind tunnel employed to perform the current study is located at the Mie University in Japan, which is illustrated in Fig. 5. The wind tunnel has an outlet diameter of 3.6 m with the air corrector size of 4.5 m × 4.5 m, and the length of test section is 4.5 m. The wind tunnel can be adjusted wind velocities in the range of 0–30 m/s. The turbulence intensity measured is less than 0.5% and the flow uniformity is less than1.2% in the mainstream direction, when the experiment is performed at mainstream wind velocity of 8.0 m/s.

Fig. 6 shows the wind turbine prototype mounted within the wind tunnel cross-section in this research. The wind turbine itself is an H-type design consisting of straight blades 1.2 m in length mounted at a radius of 1.0 m. The airfoil profile is a symmetric NACA 0021 (Fig. 7) with a chord length of 0.265 m, and number of blades is tested from 2 to 5. The coordinate system is defined for the measurements, in which the *x*-, the *y*- and the *z*-axes are set in the mainstream, the lateral and the vertical directions, respectively. The origin is set at the center height location of rotor.

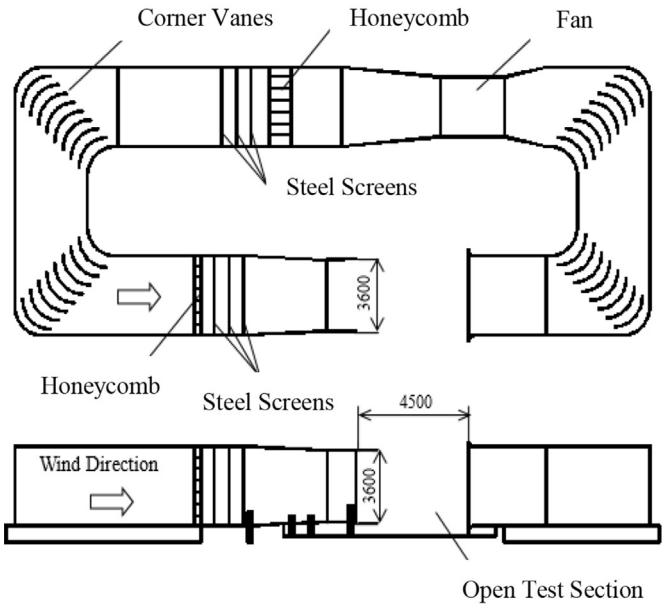
At present study, as shown in Fig. 6, in order to measure the rotor rotational speed and rotor torque, torque meter is installed in rotor shaft. The rotational speed is ranged approximately from 0 to 8000 rpm and the moment of inertia is 5.00×10^{-5} kg m^2. The capacity of torque sensor is 20 N m, minimum resolution is 10 mN m and accuracy is ±0.2% at full scale. The rotation direction of rotor is clockwise when viewed from top side of wind turbine.

2.3. Multiport pressure devices

Fig. 8 depicts the pressure measurement tube and multiport pressure measurement devices. In order to measure pressure



(a) Large wind tunnel laboratory outside view



(b) Large wind tunnel in two-dimensional sketch

Fig. 5. Large wind tunnel in this research. The wind tunnel has an outlet diameter of 3.6 m and its maximum wind velocity reaches 30 m/s.

distribution *P* applied to the rotor blade surface, in the span of airfoil center height, one of rotor blades is equipped with a total of 32 pressure taps with diameter of 0.4 mm. The measurement taps can be perpendicularly drilled to blade surface and arrangement of measurement taps are denser near the leading edge with sharp pressure, which are shown in Fig. 7. Because pressure taps could not be formed near the trailing edge, the pressure values near the trailing edge are estimated by pressure gradient at the chord length of 0.907. These points are predicted the changes of pressure gradients. The pressure taps are shifted chordwise in order to avoid interference from the other pressure taps on the airfoil surface.

The type of multiport pressure devices is ZOC22B module which incorporates 32 individual piezoresistive pressure sensors. Each block of 32 sensors has its own individual calibration valve. The ZOC22B is powered by ±15Vdc and pressures will not exceed 344.75 kPa. The sampling frequency is 1250 Hz, the total number of samples is 10,000 and the accuracy is ±0.1%. ZOC22B modules are designed to function when used with RAD3200.

The pressure on the rotor blade surface is captured by pressure sensor which is installed in the upper rotation axis, and then

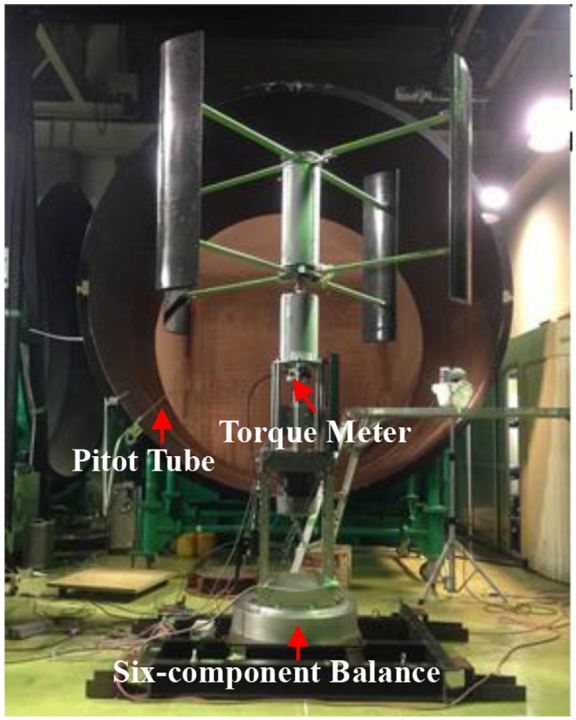


Fig. 6. Photo of test wind turbine.

transmitted through copper pipe and vacuum pipe by high-speed multiport pressure measuring device. The detecting pressure signals are transmitted to the stationary system through wireless LAN. The output signal is stored in a *PC* after passing through an *A/D* converter. The measurement pressure data are averaged at each azimuth angle of 5° BIN during one rotation. In order to carry out quantitative analysis, the pressure coefficient C_p is defined by Refs. [17,28]:

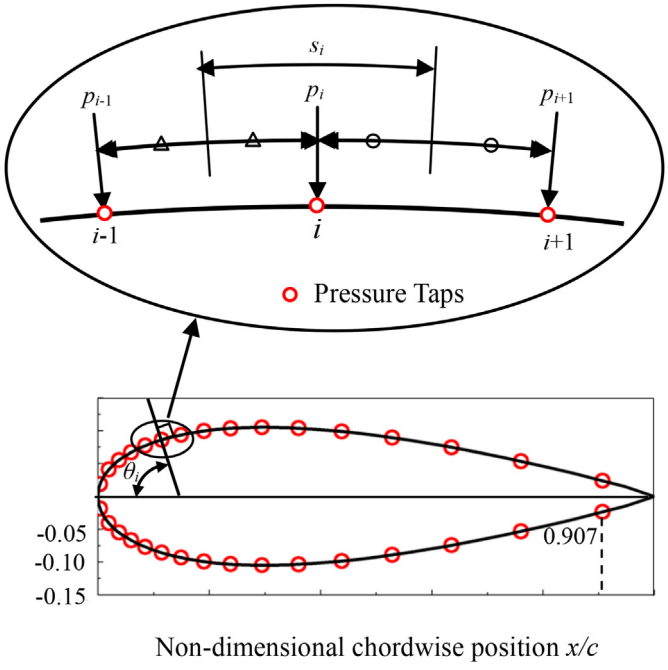


Fig. 7. The position of pressure measurement taps.

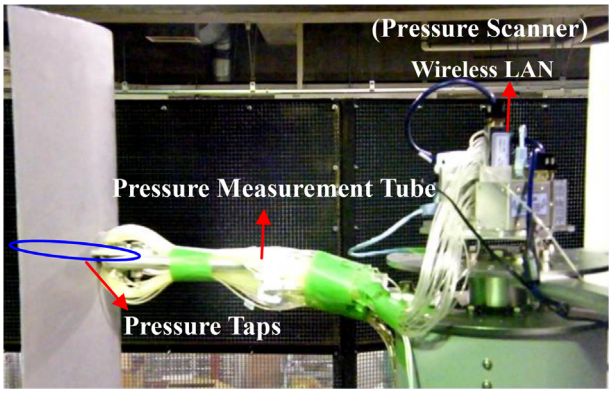


Fig. 8. Multiport pressure measurement devices. The multiport pressure measurement device is connected to pressure taps in single rotor surface by urethane tube.

$$C_p = \frac{P}{0.5\rho U_0^2} \tag{1}$$

where, P is the pressure of blade surface, ρ is the density of air and U_0 is mainstream wind velocity.

The tangential and normal forces acting on the blade can be calculated by integrating the pressure distribution along the blade surface from the following expressions [3]:

$$F_N = \sum (p_i - p_0) \cdot s_i \cdot \sin\theta_i \tag{2}$$

$$F_T = \sum (p_i - p_0) \cdot s_i \cdot \cos\theta_i \tag{3}$$

where p_i is the pressure of measurement tap when the blades are rotating and p_0 is static pressure when the blades are static. The s_i is measurement taps distance connecting the midpoint of the pressure measurement tap adjacent to each other in the airfoil section, and the θ_i the inclination angle of the pressure measurement taps in the position of i.

As shown in Fig. 8, the reference pressure of the multiport pressure measurement device is detected from the rotational center. Therefore, the values of $p_{_measure}$ by multiport pressure measurement device show lower values than the pressure p of the blade surface because of the centrifugal force acting on the air in the pressure tubing. To correct for this effect, it is determined the blade surface pressure by the following equation.

$$p = p_{_measure} + \frac{1}{2}\rho l^2 \omega^2 \tag{4}$$

where, l indicates the distance to the rotation center, which is from reference pressure sensing position to pressure taps.

Since the multiport pressure measurement device is connected to pressure taps in the rotor surface by urethane tube, so that the pressure amplitude obtained from the pressure taps will be changed and the time will also be delayed. Temporal response delay and amplitude attenuation of the pressure waveform are primarily depend on the length, internal diameter of the pressure tubing and frequency of the pressure waveforms. Therefore, in this experiment, the response of pressure taps is assayed as follows.

Experimental apparatus shown in Fig. 9 will be used to identify the response of pressure taps. The test response of pressure taps to give pressure fluctuations with some frequency components in the pressure taps of rotor surface. When the cylinder is coupled to the engine to take piston movement, pressure

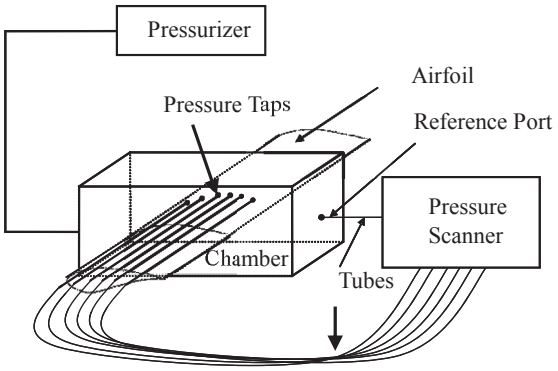


Fig. 9. Calibration of dynamic response in air of tubing.

fluctuations will be generated, and then transmitted into the coverage container of pressure holes by urethane tube of copper pipe and urethane tube with the length of 10 mm. The response of pressure taps is determined through measuring the relative reference pressure fluctuations which are changed by the attenuation of phase delay and amplitude attenuation of the pressure waveform at each pressure taps. In this study, the attenuation of phase delay and amplitude attenuation of pressure tubing is calibrated.

Due to the complexity of setup and flow, impact of measurement uncertainty on load estimation is also considered. This uncertainty is generated by the randomness of the flow and uncertainty of estimation of the average flow. To quantitatively evaluate the changes of flow around the wind turbine, local wind velocity without wind turbine is measured with LDV system [25]. The velocity data from LDV system is used to test the feasibility of estimating the loads on a VAWT aerofoil. Moreover, the uncertainty resulting from the measurement instruments can easily be overcome by a large number of samples, because a lot of precision instruments are used in this experiment.

2.4. Six-component balance

The six-component balance is installed on the basement of wind turbine. It can be measured the force and moment applied to the entire wind turbine in the three directions of x, y and z-axis. The azimuth angle is calculated based on a reference signal (one pulse/rotation) of the optical sensor output. The same to pressure measurement data, the forces data are averaged at each azimuth angle of 5° during one rotation. Voltage signal is amplified by interference compensation for distortion amplifier. The thrust coefficient C_{thrust} is shown as following [33].

$$C_{\text{thrust}} = \frac{F_x}{0.5\rho U_0^2 A} \tag{5}$$

where F_x is the thrust force and A is the swept area of wind turbine.

3. Experimental methods

3.1. Power coefficient measurement

Wind turbine rotor performance is characterized by its torque coefficient, C_Q, and power coefficient, C_{power} [3,22,29]:

$$C_Q = \frac{Q}{0.5\rho A U_0^2 R} = \frac{Q}{0.5\rho D H U_0^2 R} \tag{6}$$

$$C_{\text{power}} = \frac{P}{0.5\rho A U_0^3} = \frac{Q\omega}{0.5\rho D H U_0^3} \tag{7}$$

where Q is the torque, R is the rotor radius and D is the rotor diameter of wind turbine. H is the height of blade and ω is angular velocity of rotor. The power coefficient represents the aerodynamic efficiency of wind turbine and a function of the tip speed ratio, λ which is defined as [8,16–19,31].

$$\lambda = \frac{R\omega}{U_0} \tag{8}$$

For each test case, wind tunnel experiments are carried out under the conditions that the blade pitch angles are the optimum value at different numbers of blades, for which wind turbine has the highest power output. After that, pressure acting on the blade surface will be measured under the conditions that tip speed ratios and blade pitch angles are the optimum values at numbers of blades for two, three, four and five.

3.2. Data processing method

A better understanding of the effects of different parameters is essential in the design process. The individual parameters and their effects are not completely independent of each other but rather interlinked and complicated. The main disadvantages of a VAWT generally create drag forces when rotating into the wind [34].

Fig. 10 represents the section of VAWT wind turbine rotating clockwise and the main forces relative to the airfoil chord line. The tangential force F_T is basically the difference between the tangential components of lift and drag forces. From these relations, in order to quantitatively analyze the performance of VAWT, tangential coefficient C_T, normal coefficient C_N, lift coefficient C_L and drag coefficient C_D are shown as following, respectively [3,5].

$$C_T = \frac{F_T}{0.5\rho c U_0^2} \tag{9}$$

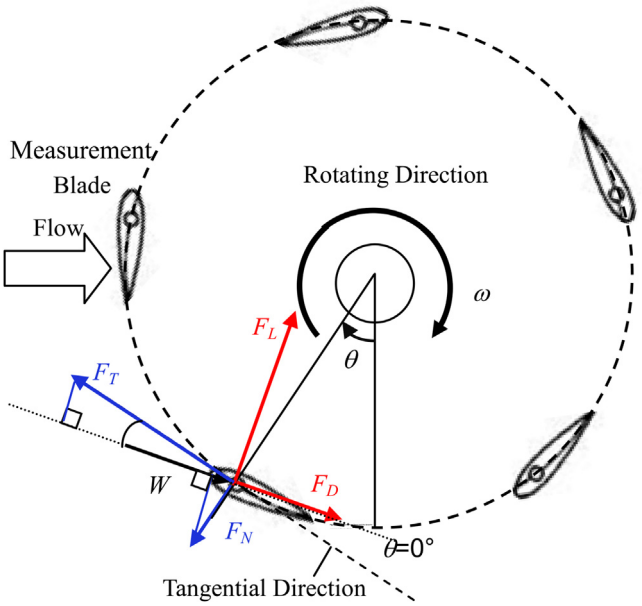


Fig. 10. Definition of forces acting on blade. θ is specified as a positive direction from 0° and the same direction as rotating direction.

$$C_N = \frac{F_N}{0.5\rho c U_0^2} \tag{10}$$

$$C_L = \frac{F_L}{0.5\rho c U_0^2} \tag{11}$$

$$C_D = \frac{F_D}{0.5\rho c U_0^2} \tag{12}$$

where F_T is the tangential force, F_N is the normal force, F_L is the lift force and F_D is the drag force per unit length.

4. Experimental results and discussion

4.1. Power coefficient curve

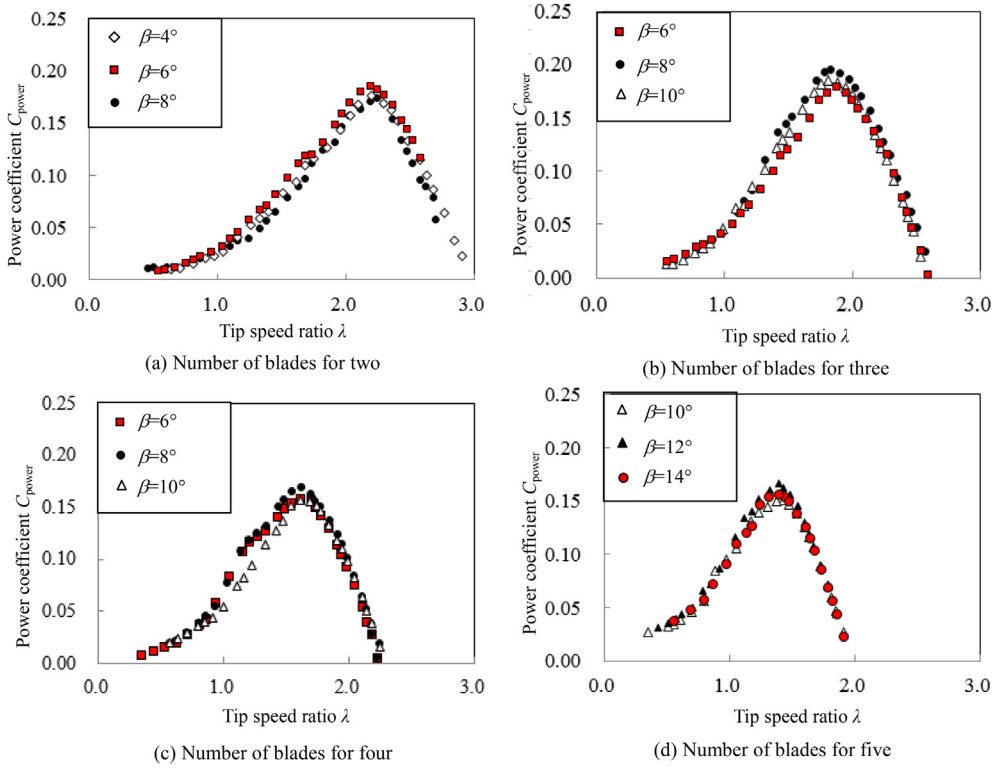
Considering the situation of wind velocity in the local area, in this research, the mainstream wind velocity is set on 8.0 m/s. To study the starting characteristic of this VAWT, by analyzing the result of six-component balance, the power coefficients against the tip speed ratio λ for different pitch angles are shown. And then, the performance of the optimized pitch angle has been analyzed at different numbers of blades.

The effect of pitch angle is shown in Fig. 11 at different numbers of blades for $N = 2$, 3, 4 and 5, respectively. Horizontal axis shows the tip speed ratio λ and vertical axis is the power coefficient C_{power}. From Fig. 11(b), the optimum power coefficient of 0.189 is obtained at $\lambda = 1.78$, $\beta = 8^\circ$ when the numbers of blades is three. It can be also seen that the power coefficient increases in the range of $0<\lambda < 1.78$ and then rapidly decreases from the optimum tip speed ratio of $\lambda = 1.78$. There is very little variation between the three cases in the pitch angle from 6° to 10°. Moreover, it can be concluded the performance decrease at a lower pitch angle (below 8°). The performance also decreases at a larger pitch angle (above 8°). In this experiment, the highest values of power coefficient are $C_{power} = 0.182$, 0.189 and 0.178, for $\beta = 6^\circ$, 8° and 10°, respectively. Meanwhile, from Figs. (a ~ d), the fluctuation of power coefficient C_{power} is slight change at different pitch angle for the same number of blades. However, the result optimum pitch angle shows the different values of $\beta = 6^\circ$, 8°, 8° and 12°, when the numbers of blades are $N = 2$, 3, 4 and 5, respectively.

Fig. 12 illustrates the power coefficient distribution at the different numbers of blades for $N = 2$, 3, 4 and 5, respectively. The pitch angle is the optimum angle which has the maximum values of power coefficient. As seen from Fig. 12, the power coefficient slowly rises in the range of $0 < \lambda < 2.19$ and then rapidly decreases from $\lambda = 2.19$ when the number of blades is two. The same above conclusions can also be obtained at other number of blades. The optimum power coefficients are 0.208, 0.195, 0.189 and 0.176, when the numbers of blades are $N = 2$, 3, 4 and 5, respectively. It is significant the starting characteristic of this wind turbine becomes better with increase of numbers of blades. The starting tip speed ratios are 0.696, 0.557 and 0.436 and 0.345, when the numbers of blades are $N = 2$, 3, 4 and 5, corresponding to Reynolds number $Re = 2.89 \times 10^5$, 2.31×10^5, 2.17×10^5 and 1.85×10^5, respectively. Therefore, the efficiency of the turbine varies greatly with wind speed, a disadvantage of constant speed operation, but it should be designed such that the maximum efficiencies are achieved at wind velocities where there is the most energy available.

4.2. Pressure distribution on blade surface

The pressure acting on the blade surface for the optimum tip speed ratios is analyzed in this section. Fig. 13(a), (b), (c) and (d)



(a) Number of blades for two

(b) Number of blades for three

(c) Number of blades for four

(d) Number of blades for five

Fig. 11. Power coefficient for different pitch angles. The number of blades is from two to five. The horizontal axis shows the tip speed ratio λ.

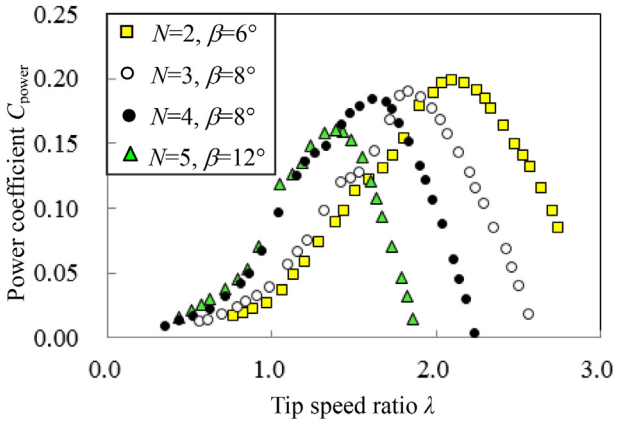


Fig. 12. Power coefficient for different numbers of blade. The pitch angle is the optimum angle and the number of blades is from two to five.

indicate the pressure distribution at the different numbers of blades for $N = 2$, 3, 4 and 5, respectively. The distribution of pressure coefficients are shown when rotor blade passes through the azimuth angles of 0°, 90°, 180° and 270°. The pressure difference between the upper and lower surfaces of the airfoil is determined by the following parameters: characteristics of airfoil section, number of blades and azimuth angle. Horizontal axis shows the non-dimensional chord station *x/c* and vertical axis is pressure coefficient non-dimensionalized by the dynamic pressure of undisturbed flow.

From Fig. 13(a) to (d), it is noted that the pressure distribution acting on the blade surface is much fluctuated with the change of azimuth angle. Suction steadily increased from the trailing edge region with chordwise position for different numbers of blades and up to a maximum value around the leading edge. The pressure difference is increased when rotor blade passes through the upstream region. This result indicates that a large fluid force is occurred at upstream region. Here, the rotational force acting on single blade can be calculated by integrating the pressure distribution, which will be discussed along with disparity in the next section.

Fig. 13(d) illustrates the fluctuations of pressure distribution at number of blades for $N = 5$. Pressure distribution on suction surface shows enough negative value except 180° case. At 180°, the blade moves to downwind direction, so the resultant flow velocity relative to blade becomes minimum so that the pressure difference becomes almost zero because the blade moves as the same direction of the mainstream flow. The other case is that the blade produces the lift force with enough relative speed and angle of attack. The maximum value of pressure coefficient depends on azimuth angle. The stagnation pressure for 90° is larger than that for 270°. In contrast, when the blade passes through the downstream region of 270 degrees-azimuth, pressure difference becomes smaller due to the balance of the angle of attack and the velocity deficit in the wake.

As can be seen in Fig. 13, it seems that the pressure difference tends to decrease with increase of numbers of blades. According to Fig. 13(a ~ d), the same tendency of pressure distribution are also found at the other numbers of blades. However, unlike the (a ~ c), the maximum value of pressure difference for the number of blades for five can be seen in 0 degrees-azimuth. It seems that the main effects due to balance of the angle of attack and velocity deficit by the wake which has the smallest value at the number of blades for five.

4.3. Tangential and normal forces for single blade

The tangential and normal forces acting on single blade cause the bending moment and the twist of rotor shaft. The tangential and normal forces can be calculated by integrating the pressure distribution. The tangential and normal force coefficients are non-

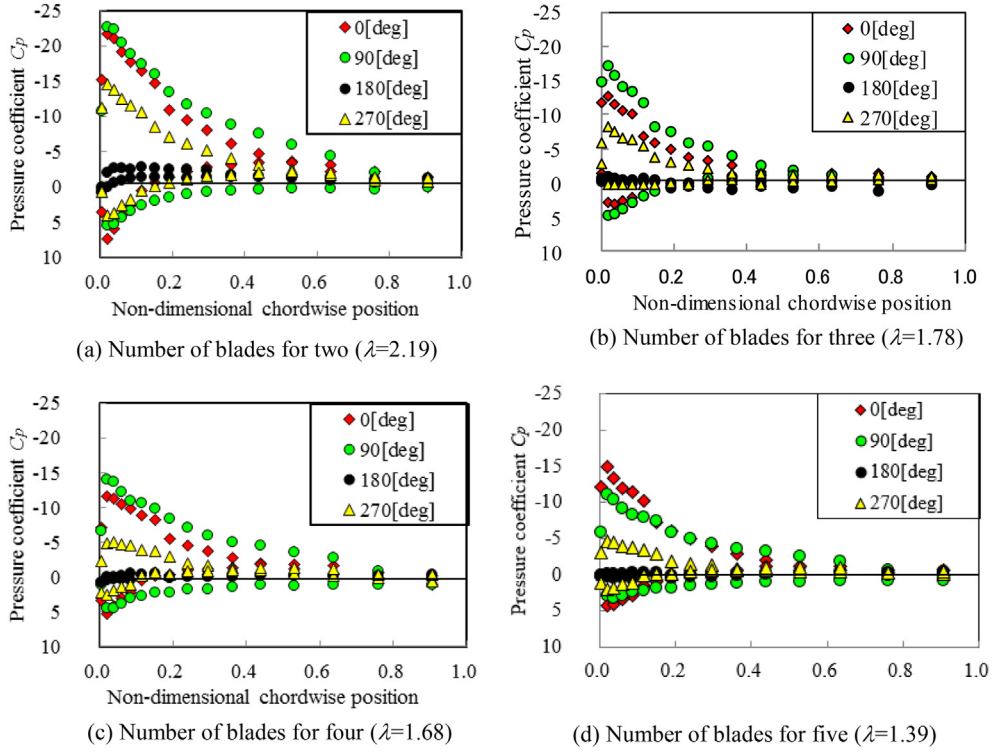


Fig. 13. Pressure distribution for different numbers of blades when the rotor blade passes through the azimuth angles of 0°, 90°, 180° and 270°. The number of blades is from two to five.

dimensionalized by the airfoil chord length c and the dynamic pressure in the mainstream wind velocity of $U_0 = 8.0$ m/s.

Fig. 14 represents the evolution of the tangential coefficient against azimuth angle for different numbers of blades. From this figure, it is apparent that when the blade is moving to upstream region around azimuth angle 100°, the tangential force coefficient C_T takes a maximum value. On the contrary, the tangential force coefficient becomes small and smooth when azimuth angle is at the region of 180 to 45 degrees-azimuth. The reason for this tendency is mainly considered to be the influence of angle of attack α, which becomes larger at upstream region and smaller at downstream region. While the agreement between different numbers of blades is quite good on the upstream region where aerodynamic loading is larger, some discrepancies can be seen at the position of azimuth angle during rotation. The peak value of tangential coefficient decreases with the increase of number of blades.

Normal force coefficients C_N acting on the blade as a function of azimuth angle at different numbers of blades are shown in Fig. 15. In contrast with C_T in Fig. 14, the fluctuations increases with the increase of number of blades in the upstream region of azimuth angle of $\theta = 0°$~$180°$. This tendency of little change in the downstream region indicates a significant effect of the wake behind the turbine. Furthermore, the fluctuation amplitudes of normal force coefficients reach the largest absolute value around the upstream region of 90 degrees-azimuth. The maximum values are about −8.544, −8.273, −5.168 and −4.278, when the numbers of blades are $N = 2$, 3, 4 and 5, respectively.

4.4. Lift and drag for single blade

In Fig. 16, the estimated results of lift coefficients C_L for different numbers of blades are plotted against angle of attack α. Horizontal axis shows the angle of attack α and vertical axis is lift coefficient C_L. In this figure, the characteristic curves of lift against α form asymmetric hysteresis loops and show clearly different features. When the numbers of blades are $N = 2$, 3, 4 and 5, the stall happens at angle of attack about α = 13.07°, 12.87, 12.18° and 11.36°, respectively. Along with the increase of number of blades, the maximum lift coefficients decrease and reduce to 8.86, 8.12, 6.97 and 5.76, respectively. From these situations, flow is separated from the airfoil surface and lift coefficient decreases sharply.

Fig. 17 illustrates the fluctuation of drag coefficient C_D against angle of attack α at different numbers of blades. From this figure it is noted the same phenomena as discussed for lift coefficient:

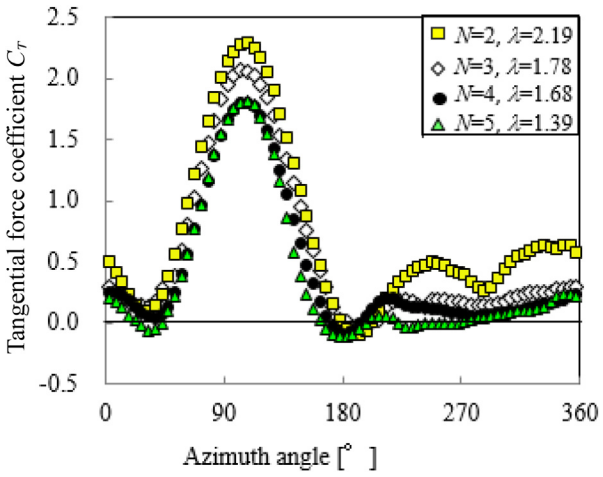


Fig. 14. Tangential force coefficient for different numbers of blades. The number of blades is from two to five.

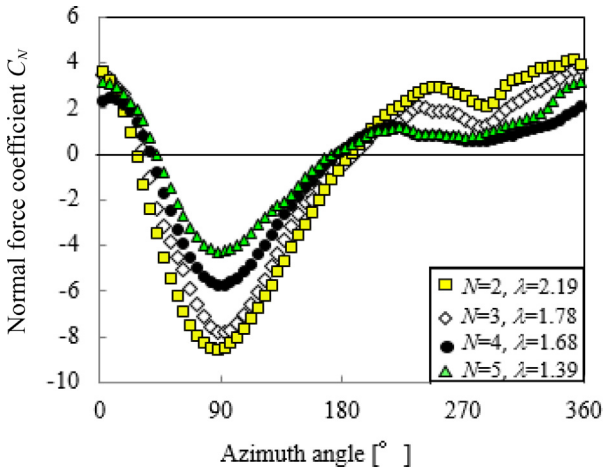


Fig. 15. Normal force coefficient for different numbers of blades. The number of blades is from two to five.

characteristic curves of drag against α form asymmetric hysteresis loops. Moreover, the results suggest the maximum value of drag coefficient decreases with the increase of number of blades. When the numbers of blade are $N = 2$, 3, 4 and 5, the maximum values are about 1.06, 0.96, 0.93 and 0.85, respectively. For higher number of blades, the maximum value of C_D rises further, though the fluctuations of C_D become narrow.

4.5. Fluctuation of power and torque coefficients

In this section, for the purposes of investigating the fluctuation of power and torque coefficients depending on azimuth angle, the total power coefficient is divided into the upstream and downstream region. Finally, the experimental results estimated from the measured values are compared between pressure distribution and torque meter.

Fig. 18 shows the fluctuation of power coefficient C_{power} (for single blade) against azimuth angle at different numbers of blades. When the rotor blade is moving of azimuth angle from about 45° at the upstream region, the power coefficient C_{power} becomes larger and power coefficients reach their maximum values at the azimuth angle of 100°. In contrast, when the wind turbine passes through the downstream region, the power coefficients are reduced.

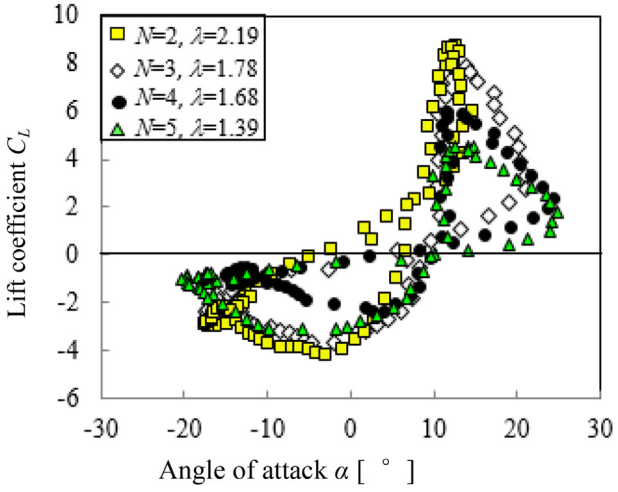


Fig. 16. Lift coefficient for different numbers of blades. The number of blades is from two to five.

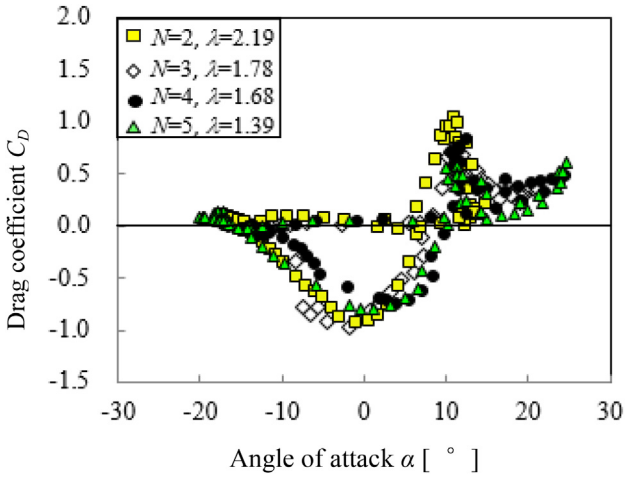


Fig. 17. Drag coefficient for different numbers of blades. The number of blades is from two to five.

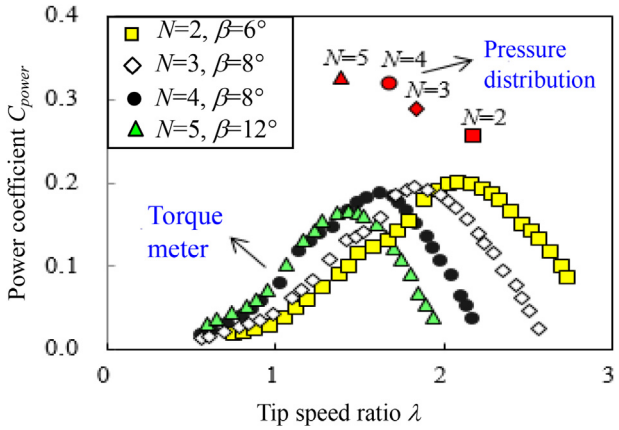


Fig. 19. Fluctuation of power coefficient for the whole turbine at different numbers of blades. The red points represent the values estimated by pressure distribution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Meanwhile, it also must be noted that in the cases of numbers of blades for four and five, the waveforms of power coefficients C_{power} have similar characteristics. When the numbers of blades are $N = 2$, 3, 4 and 5, the maximum power coefficients are about $C_{power} = 0.410$, 0.387, 0.338 and 0.326, respectively.

Fig. 19 depicts the power coefficients C_{power} (for the whole turbine) as a function of the tip speed ratio for six-component balance at different numbers of blades. As can be clearly seen, the peak of power coefficient decreases with the increase of number of blades. When the numbers of blades are $N = 2$, 3, 4 and 5, the maximum power coefficients are about $C_{power} = 0.198$, 0.192, 0.189 and 0.173, respectively. However, the optimum tip speed ratio is expected to be increased with the increase of number of blades.

Fig. 20 compares the fluctuation of torque coefficients C_Q (for the whole turbine) as a function of tip speed ratio for six-component balance at different numbers of blades. As noted in the figure, the similar tendency to C_{power}, the value of C_Q slowly rises before reaching the optimum tip speed ratio and then rapidly decreases from optimum tip speed ratio. Moreover, an important difference to C_{power} is that the values of C_Q is expected to be increased with the increase of number of blades. This will be due to the ratio of the lift to drag on the two bladed VAWT being proportionally higher than the other bladed design and thus generating higher torque and higher rotational speeds for a given tip speed ratio.

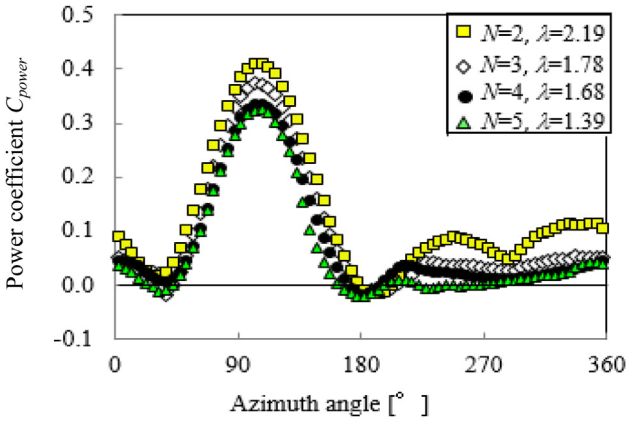


Fig. 18. Fluctuation of power coefficient for single blade at different numbers of blades. The number of blades is from two to five.

With two-bladed VAWT as an example, the comparisons between wind tunnel measurements and CFD calculations are shown in Fig. 21. The black values are obtained from wind tunnel measurements. The blue and red values represent the CFD calculation with original and velocity correction in 2D, respectively. The yellow value is obtained from pressure distribution at the optimum tip speed ratio. From this figure, the value from pressure distribution indicates a good agreement with the value from 2D CFD calculations (Velocity Correction) and also shows a larger value than torque meter. Therefore, it is very reasonable of the values obtained from the pressure and experimental method is not problematic.

The red points which are shown in Figs. 19 and 20 represent the values estimated by the pressure of blade surface convey larger value than torque meter, especially in the number of blades for five. It seems that the difference in the power coefficient between is the effect of the blade tips and the additional loss from the support structure. Moreover, another reason for this discrepancy is probably the mechanical loss of the rotating system, for example, the loss due to the friction between the turbine shaft and the two bearings and due to the elastic torsion produced by flexible couplings. It is still noteworthy that the result from pressure distribution only considered the center of blade span, not take into

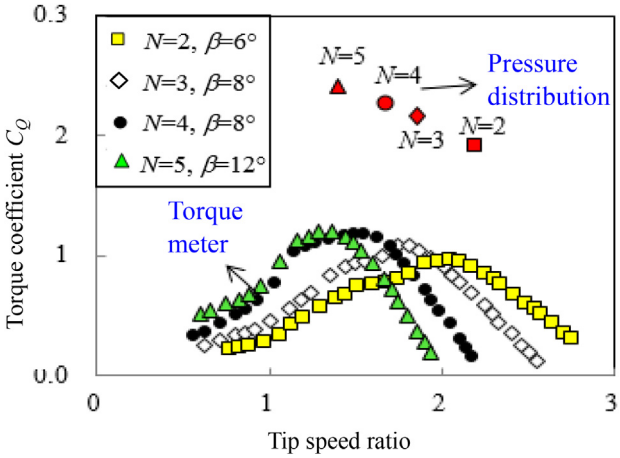


Fig. 20. Fluctuation of torque coefficient for whole turbine at different numbers of blades. The red points represent the values estimated by pressure distribution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

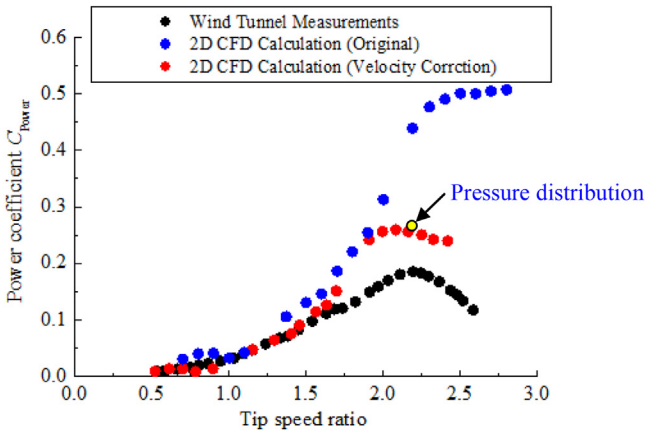


Fig. 21. The comparisons between wind tunnel measurements and CFD calculations. The black values are obtained from wind tunnel measurements. The blue and red values represent the CFD calculation with original and velocity correction in 2D, respectively. The yellow value is obtained from pressure distribution at the optimum tip speed ratio. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

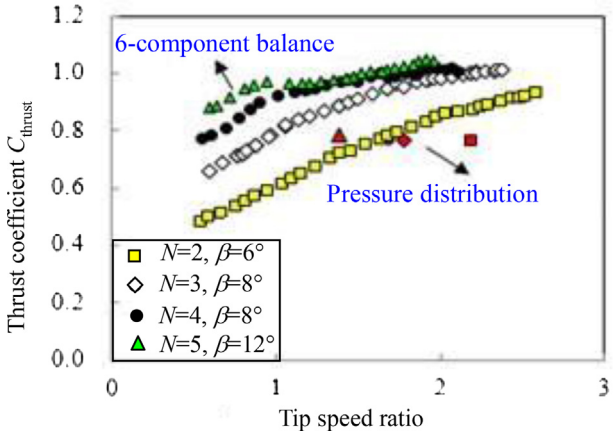


Fig. 22. Thrust coefficient for different numbers of blade. The red points represent the values estimated by pressure distribution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

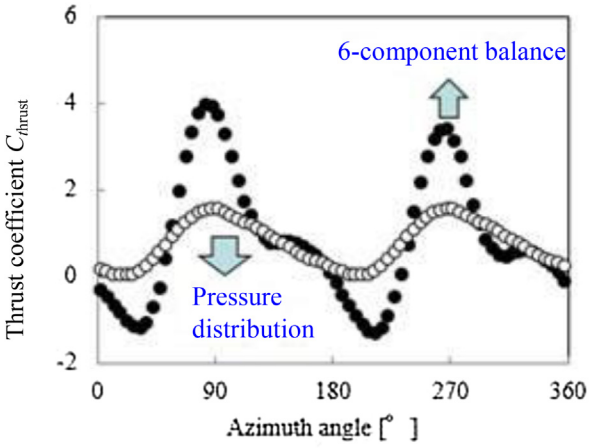


Fig. 23. Thrust coefficient for 2 blade. The black and white points represent the values estimated by 6-component balance and pressure distribution.

account pressure distribution at other positions (3D effect), especially at the position of the tip blade where tip vortices are generated. Furthermore, tip loss is the tendency for vorticity to be trailed from the tip of a wind turbine blade, and from any other point where the circulation distribution changes, thereby reducing the blade's effectiveness. This is the main factor indicated as responsible for the differences between 2D and 3D of a VAWT's flow. However, in this research, the pressure acting on the blade surface is analyzed only in 2D. These mentioned reasons will be more detail verified in our future studies by wind tunnel experimental and CFD.

In summary, the performance of wind turbine can be characterized by the manner in which the three main indicators, power, torque and vary with wind speed. Wind turbine with two blades has the maximum power coefficient and minimum torque coefficient. However, starting wind velocity is not very satisfactory and requires larger value. Therefore, in high wind velocity areas, two blades has a higher annual generating capacity. In contrast, when the number of blades is five, three is low starting torque (high torque at low tip speed ratios) which is of importance and this also allows small amounts of power to be developed at low wind speeds, ideal for trickle charging batteries. Wind turbine with five blades has a higher annual generating capacity at low wind velocity areas.

4.6. Fluctuation of thrust coefficient

In order to compare the results of six-component balance and pressure distribution for the whole wind turbine, thrust coefficients C_{thrust} against azimuth angle θ at optimum tip speed ratio are shown in Fig. 22, the fluctuation amplitudes obtained by six-component balance show larger value than the results of pressure distribution, due to the influence of resonance. Furthermore, the thrust coefficients increase along with the increase of number of blades.

Fig. 23 compares the fluctuation of thrust coefficients C_{thrust} as a function of tip speed ratio between pressure distribution and six-component balance at the number of blades for two. From this figure, it is clear that there is a periodical change of C_{thrust} which is composed of two cycles during one rotation. In addition, the fluctuation amplitudes obtained by six-component balance show larger value and more rough than the results of pressure distribution, due to the influence of resonance. As the above mentioned, the effect of the rotary friction and the additional loss from support structure are not considered in pressure measurement. On the other hand, the effect of pressure in 3D will also be analyzed in the future study.

5. Conclusions

It is very important and difficult to develop the simple design equations for straight-bladed VAWT. This is mainly due to the sensitivity of the power coefficient produced by wind turbine to the dynamic stall phenomenon and the turbulence level around the blades. Since the presence of power and aerodynamic performance for the simulated tip speed ratio were discounted in previous studies, the effect of different numbers of blades was not explicitly predicted in wind tunnel experiment. In this paper, in order to determine the parameters for development of the simple design standards for straight-bladed VAWT, the evaluation of energy performance and aerodynamic forces at different numbers of blades were investigated through directly measurement of the load acting on the blade in wind tunnel.

The experimental results of pressure distribution and six-component balance were proposed for a classical NACA 0021. Through the analysis of the distribution of the tangential force, normal force, power, torque coefficients as a function of azimuthal position for single blade, aerodynamic force characteristics were

investigated for several values of number of blades, allowing a comparison between pitch angle and optimum power coefficient.

In present study, it is found that the power which is absorbed from wind by wind turbine mainly depends on upstream region of azimuth angle of $\theta = 0°$~$180°$. Furthermore, the power coefficients decrease with the increase of number of blades. For single blade, the maximum power coefficient is up to 0.410 at 2 blade. However, the maximum power coefficient is only 0.326 at 5 blade. Moreover, the fluctuation amplitudes from six-component balance show larger value than the results of pressure distribution, due to the influence of resonance. Furthermore, in high wind velocity areas, two blades has a higher annual generating capacity. In contrast, wind turbine with five blades has a higher annual generating capacity at low wind velocity areas.

The experimental investigation presented in this paper helps characterizing the VAWT aerodynamic performance and contributes to a better understanding of the physics involved. These results are very important to develop the simple design equations and applications for straight-bladed VAWT because of the lack of experimental results. Evaluating these methods for a full 3D measurement is a very important future research field and it is being investigated by the authors.

Acknowledgment

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